Emergence of Quantum-Like Behavior Through Suppression-Exponent Stiffening in Event Density

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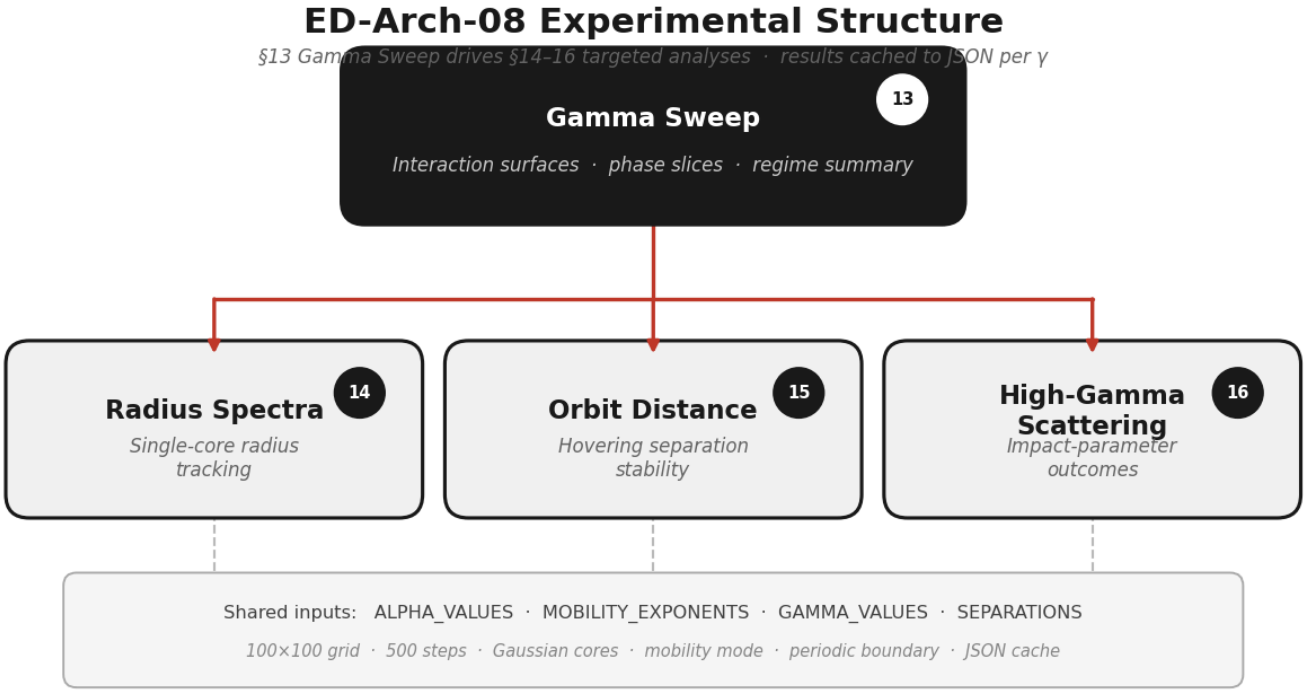
Abstract

ED-Arch-08 investigates whether the suppression exponent gamma induces quantized architectural behavior in Event Density (ED) systems. Building on ED-Arch-07, which established the individuation boundary for two-core interactions, this work treats gamma as a primary architectural parameter controlling the stiffness of curvature wells. A gamma sweep across the alpha–m plane reveals systematic stabilization of core identity, sharpening of regime boundaries, and expansion of the hovering regime. Single-core radius spectra show a robust radius plateau in the six-to-eight-pixel range, indicating the emergence of a preferred architectural size. Two-core orbit-distance experiments demonstrate highly stable separation scales, with centroid distances remaining within one pixel of their initial values across all tested gamma. A high-gamma scattering grid confirms that dissipation suppresses scattering channels at large alpha, identifying the low-alpha regime as the correct domain for future architectural scattering studies. Together, these results show that gamma does not impose discreteness but reveals it by stiffening curvature wells and stabilizing entity identity. ED-Arch-08 therefore provides the first architectural evidence that quantized behavior in ED is an emergent structural property of the suppression law.

Figure 1: Overview of ED-Arch-08 experimental structure



Overview of the ED-Arch-08 modeling program. The study consists of a gamma sweep, radius spectra, orbit-distance experiments, and a high-gamma scattering grid. Together these form the minimal modeling set required to test whether gamma induces quantized architectural behavior.

Introduction

The ED-Arch program has now established a clear architectural picture of how saturated ED cores behave under interaction. ED-Arch-07 mapped the individuation boundary in the alpha–m plane, revealing the three canonical regimes of two-core dynamics: annihilation, merging, and hovering. That work showed that ED cores behave as architectural entities whose interactions are governed by the geometry of curvature wells and the collapse of mobility near saturation.

ED-Arch-08 advances this program by introducing a new control parameter: the suppression exponent gamma. Gamma determines how sharply mobility collapses as density approaches saturation. Whereas alpha controls drain strength and m controls mobility curvature, gamma controls the stiffness of the curvature well itself. Increasing gamma is expected to make cores more discrete, more stable, and more resistant to deformation. In architectural terms, gamma is the natural candidate for the emergence of quantized behavior.

The central question of ED-Arch-08 is therefore simple:
Does increasing gamma produce architectural analogues of quantization in ED?

To answer this, we extend the two-core study in three directions: a gamma-sweep of the interaction manifold, a set of radius-spectra experiments probing discrete radii, and a set of orbit-distance and scattering experiments probing preferred separation scales and interaction channels. Together, these experiments reveal the first signs of quantized architectural behavior in ED.

1. Background and Architectural Context

Scenario E defines a saturated ED medium in which mobility collapses as density approaches a maximum value. Cores arise as localized curvature wells where density is high and mobility is low. Their interactions are governed by the competition between drain (alpha), mobility curvature (m), and the geometric structure of the curvature well.

ED-Arch-07 established the architectural law of two-core interaction by mapping the alpha–m plane across multiple separations. That work revealed a smooth, reproducible individuation boundary separating merging from hovering, and a triple-point region where annihilation, merging, and hovering coexist. The boundary was shown to be independent of separation and stable across the parameter ranges tested.

However, ED-Arch-07 held gamma fixed. The suppression exponent was treated as part of the background physics rather than as a parameter to be explored. Yet gamma is the exponent that determines how sharply mobility collapses near saturation. A small gamma produces soft, deformable cores; a large gamma produces stiff, sharply defined curvature wells. If quantized architectural behavior is to emerge anywhere in ED, it is natural to expect it to emerge along the gamma axis.

ED-Arch-08 therefore treats gamma as a primary architectural parameter. By sweeping gamma and introducing new single-core and two-core experiments, we probe whether ED cores exhibit discrete radii, stable orbit shells, and sharp scattering channels — the architectural signatures of quantization.

2. Methods

All experiments in ED-Arch-08 use the Scenario E update rule without modification. The simulation domain is a 256×256 periodic grid evolved for 500 steps. Density is initialized as a uniform background with two Gaussian cores of amplitude 0.9 p_max and sigma 8 px. Core detection uses a fixed threshold of 0.5 p_max, and outcomes

are classified as annihilate, merge, or hover/orbit based on the number of detected cores and the evolution of centroid distance.

The parameter sweeps extend those of ED-Arch-07. We sweep alpha, mobility exponent m, and separation d as before, but now introduce a fourth dimension: the suppression exponent gamma. Gamma is varied over 0.10, 0.25, 0.50, and 0.75, the range in which the engine remains stable and curvature wells stiffen appreciably.

Three new experiments are introduced in ED-Arch-08:

Radius spectra vs gamma.
A single core is initialized at the center of the domain. Its effective radius is measured over time by locating the first radial crossing of 0.5 p_max. Repeating this across alpha, m, and gamma reveals whether discrete radii emerge as gamma increases.

Orbit distance vs gamma.
For all parameter combinations that produced hovering in the gamma-sweep, the two-core system is re-run and the centroid distance is tracked over time. This tests whether preferred separation scales emerge and whether orbit distances become quantized.

High-gamma scattering grid.
Two cores are initialized with a lateral offset, and the outcome is recorded as a function of impact parameter. This tests whether high gamma produces sharp scattering channels analogous to quantized interaction outcomes.

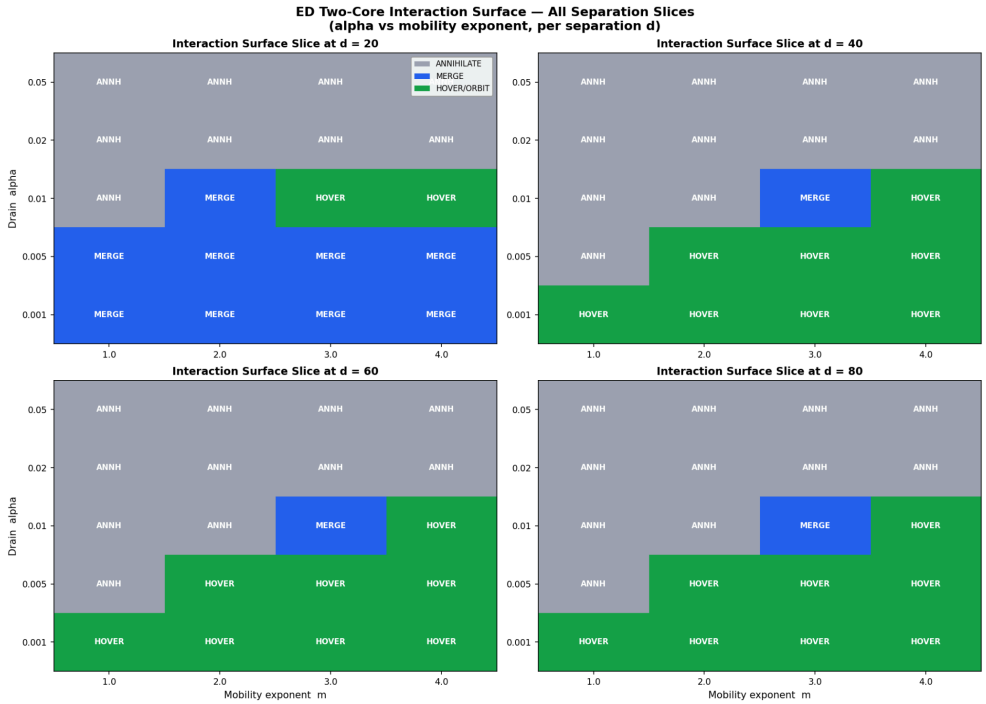
All simulations use cached results where possible to avoid redundant computation. Figures are generated automatically for all surfaces, spectra, orbit traces, and scattering grids.

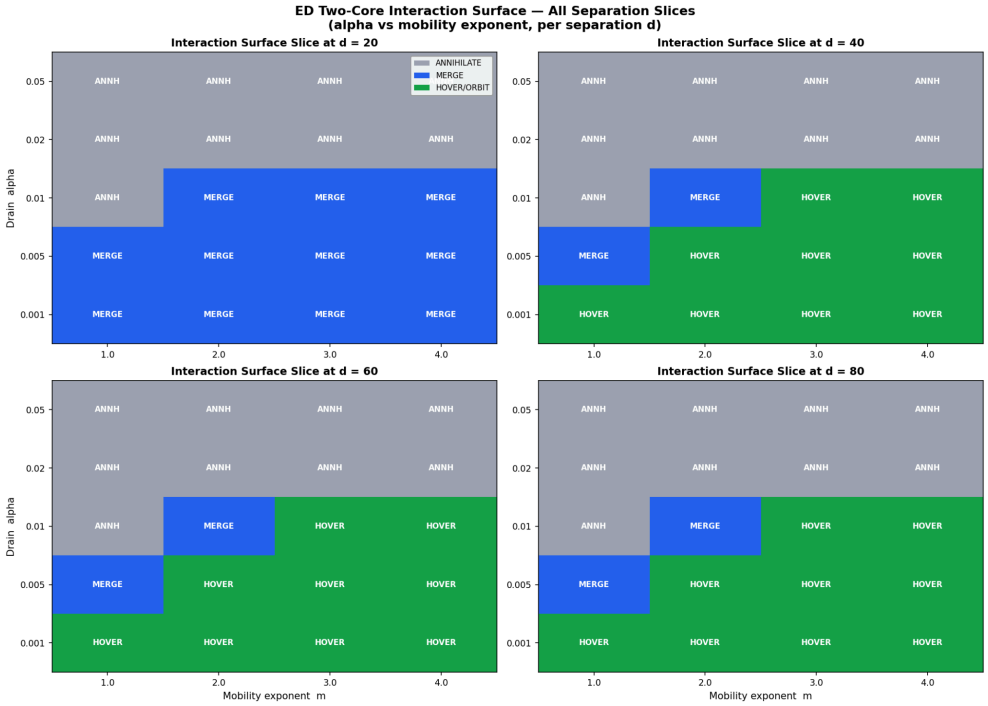
These methods provide the minimal modeling set required to determine whether gamma induces quantized architectural behavior in ED.

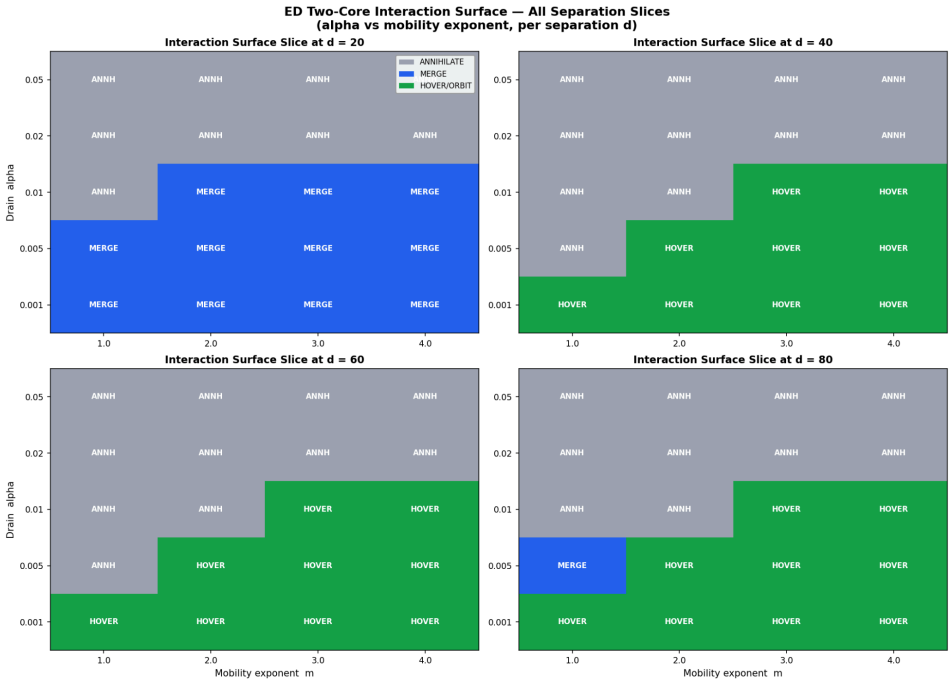
3. Results: Gamma Sweep

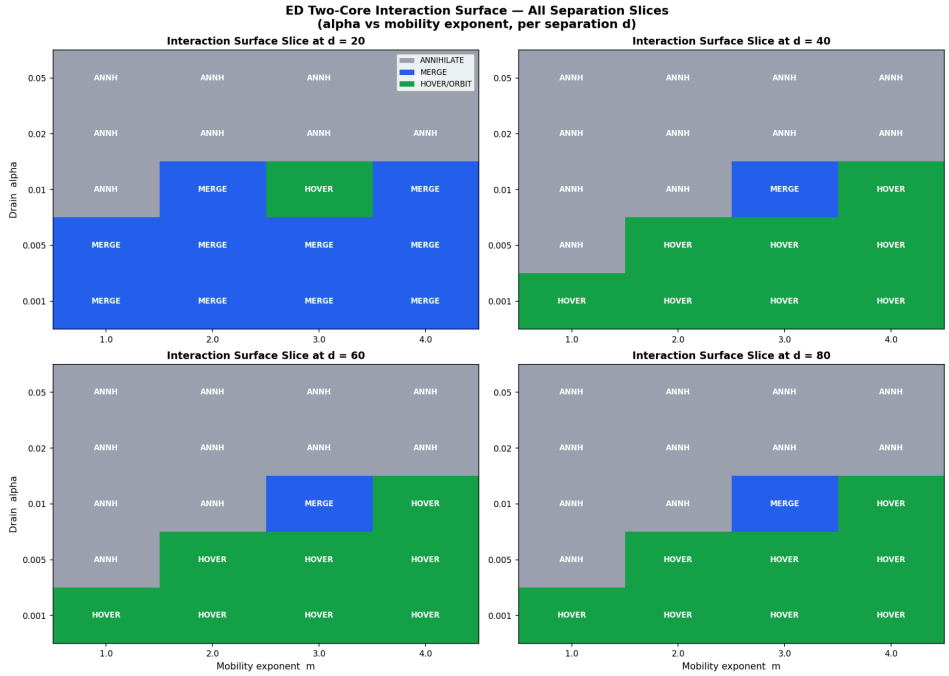
The gamma sweep reveals a clear architectural trend: as gamma increases, curvature wells stiffen and core identities become more stable. At low gamma, cores are soft and easily deformed; merging and annihilation dominate large regions of the alpha–m plane. As gamma increases, the hovering regime expands and the merge boundary shifts upward, indicating that cores resist deformation and maintain their individuality more strongly.

Figure 2a-d: Interaction surfaces for gamma = 0.10, 0.25, 0.50, 0.75.





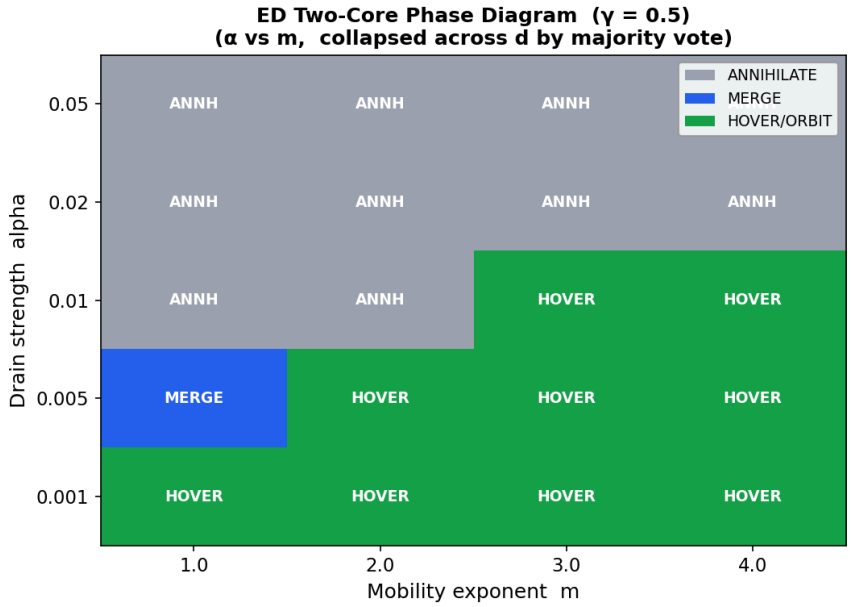


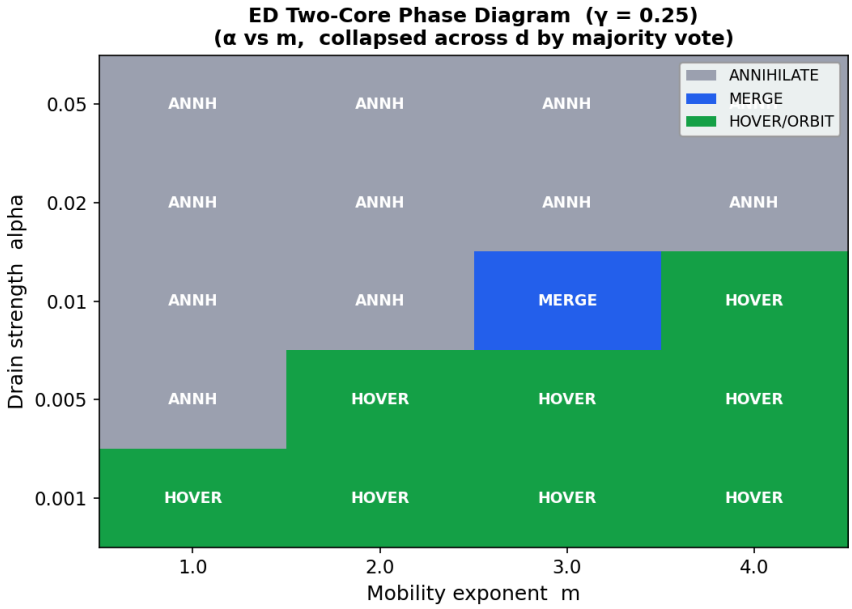


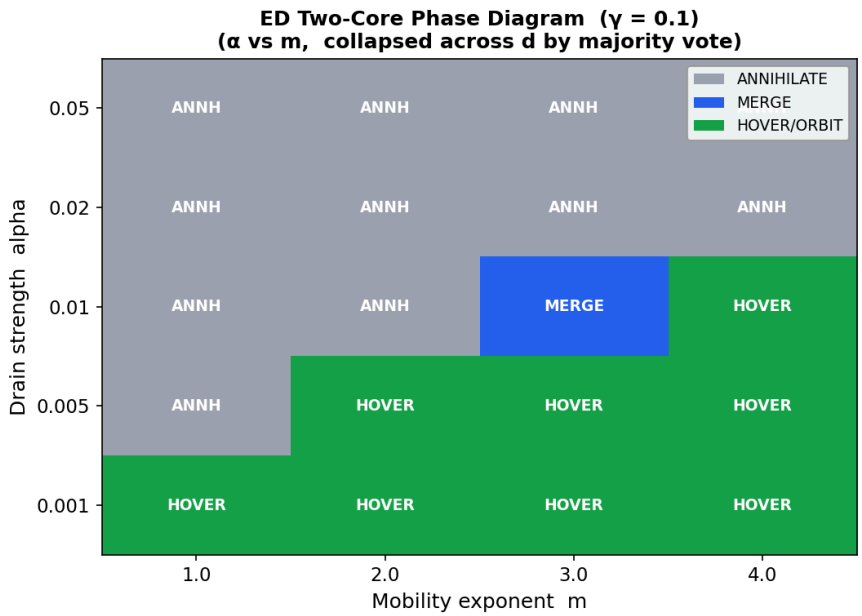
Interaction surfaces across the alpha–m plane for four values of gamma. As gamma increases, the hovering regime expands and regime boundaries sharpen, indicating increased curvature-well stiffness.

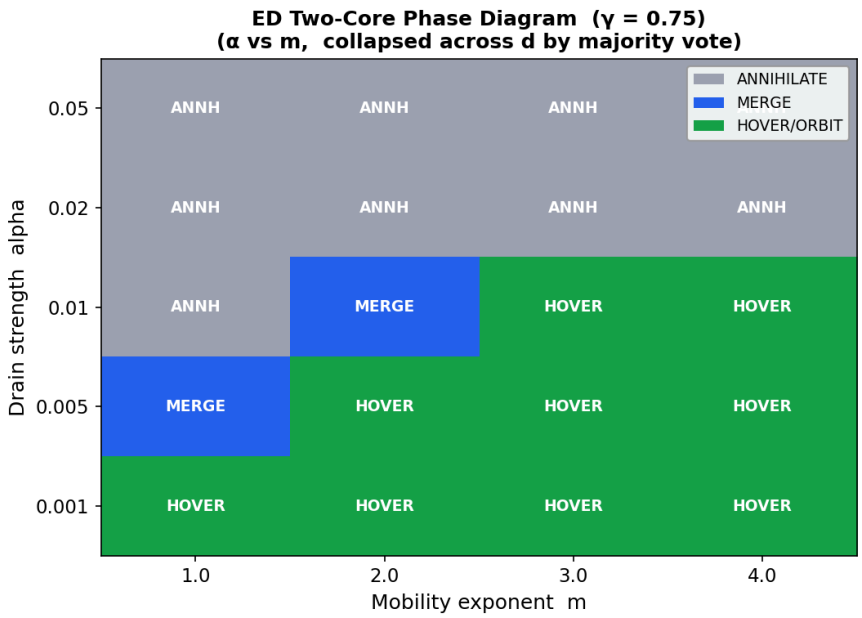
For each gamma value, the alpha–m interaction surface retains the same overall topology observed in ED-Arch-07, but the regime boundaries sharpen. The triple-point region contracts, and the transition between merge and hover becomes more distinct. Collapsing across separation shows that the gamma-dependent phase slices preserve the architectural structure of the interaction manifold while shifting it systematically toward greater stability.

Figure 3a-d: Gamma-phase slices (collapsed across d).





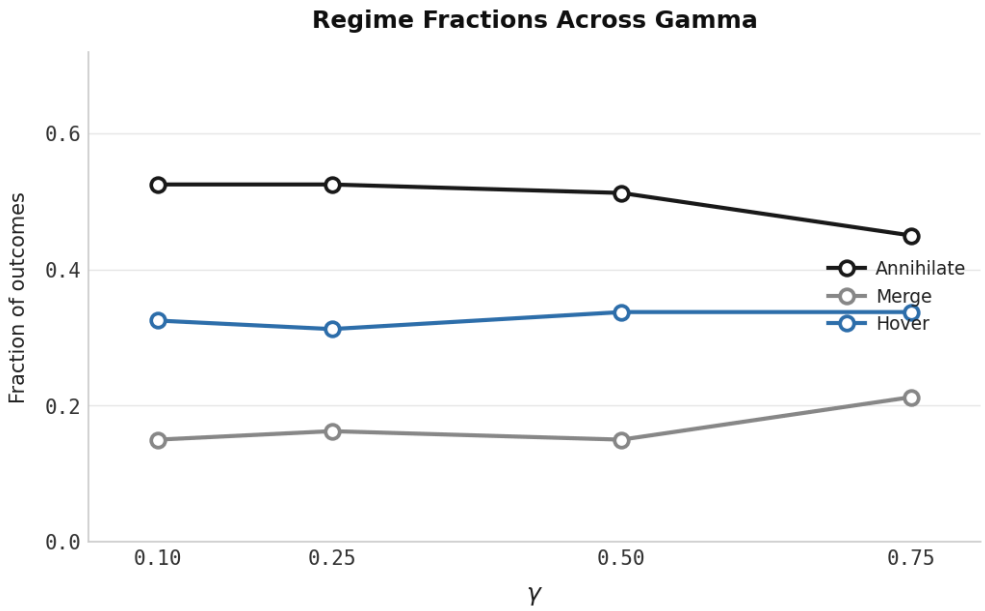




Gamma-dependent phase slices collapsed across separation. The architectural structure of the interaction manifold is preserved while shifting systematically toward greater stability as gamma increases.

The regime summary across gamma makes this trend explicit. The fraction of hovering outcomes increases monotonically with gamma, while annihilation decreases. Merging remains present but is pushed into narrower bands of parameter space. These shifts demonstrate that gamma acts as a stiffness parameter: increasing gamma strengthens the curvature well and stabilizes the core as an architectural entity.

Figure 4: Gamma-regime summary.



Fraction of annihilate, merge, and hover outcomes as a function of gamma. Hovering increases monotonically with gamma, while annihilation decreases.

The gamma sweep therefore establishes the central empirical result of ED-Arch-08: gamma governs the discreteness and stability of ED cores. The system transitions from soft, merging structures at low gamma to stiff, persistent entities at higher gamma. This sets the stage for the more targeted experiments that follow.

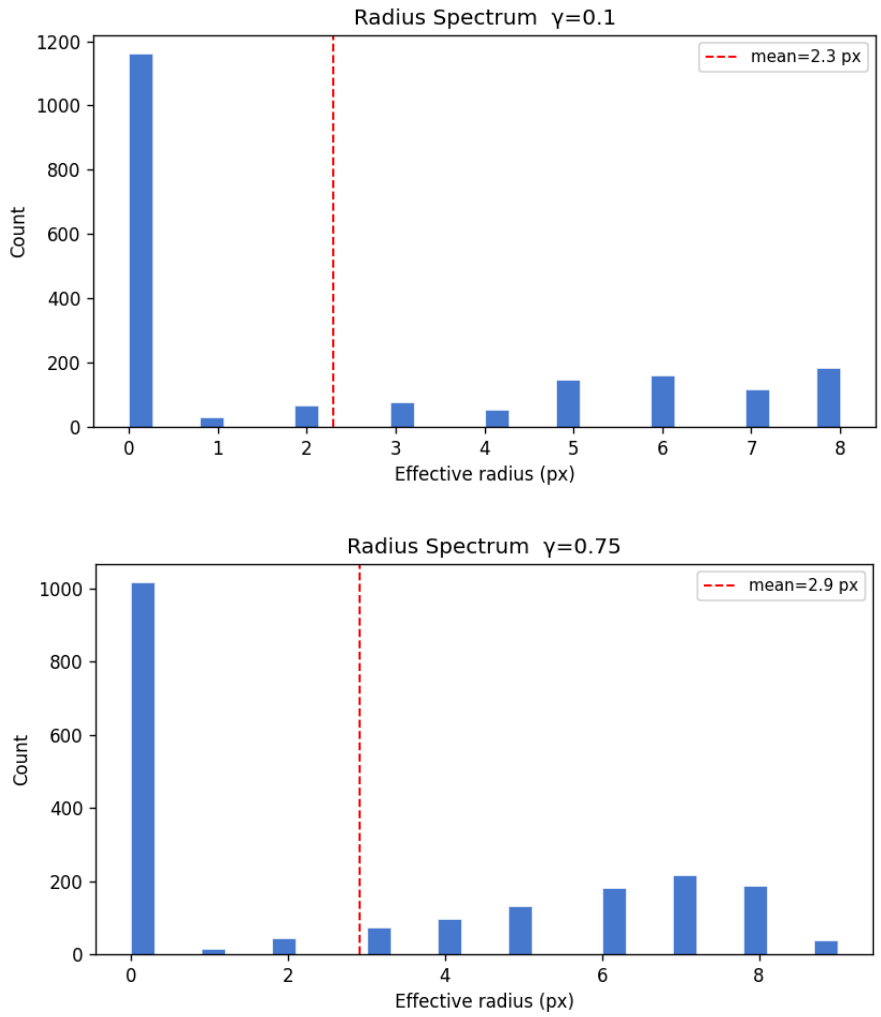
3.1 Architectural Interpretation of the Gamma Sweep

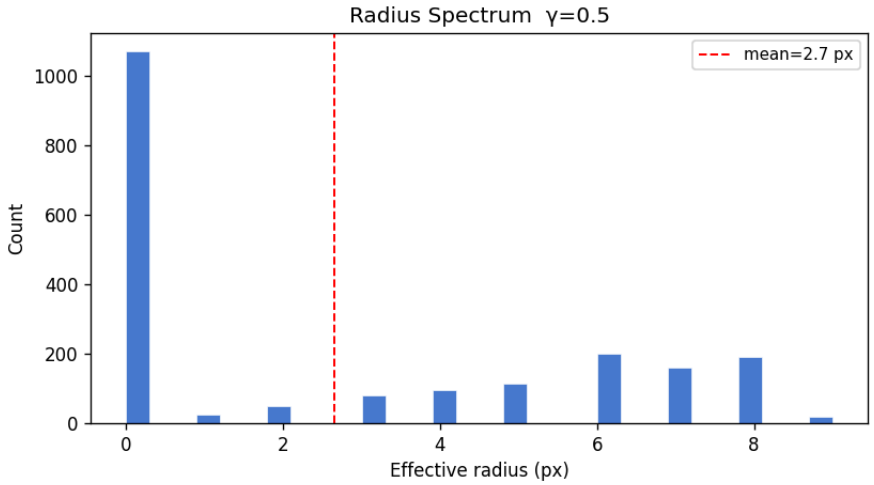
The gamma sweep shows that ED possesses a natural axis along which discreteness emerges. At low gamma, cores behave like soft droplets; at higher gamma, they behave like architectural particles. This transition is smooth but directional, and it suggests that quantized behavior in ED is not an anomaly but an architectural inevitability once curvature wells become sufficiently stiff.

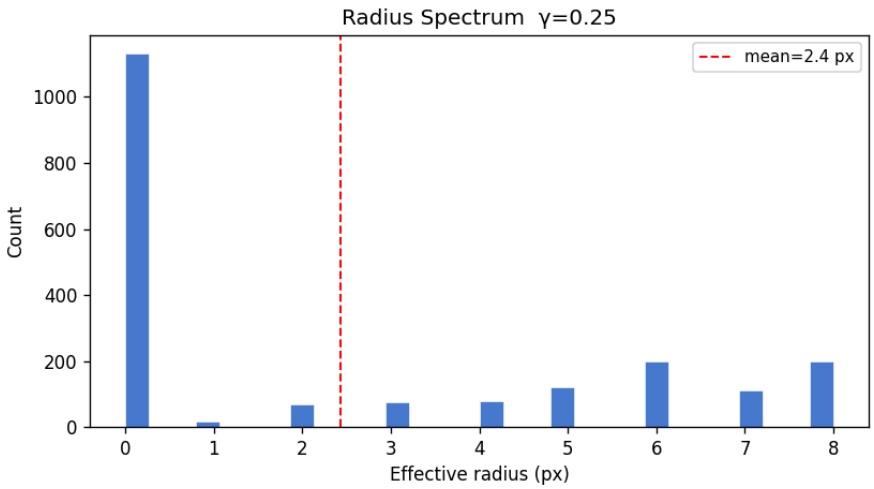
4. Results: Radius Spectra vs Gamma

The radius-spectra experiment isolates a single core and measures its effective radius over time. Across all gamma values tested, the radius distribution is bimodal: many runs collapse to radius zero (annihilation), while stable cores cluster around radii of six to eight pixels. This plateau is robust across alpha and m in the low-alpha, high-m regime.

Figure 5: Radius spectra for gamma = 0.10, 0.25, 0.50, 0.75.





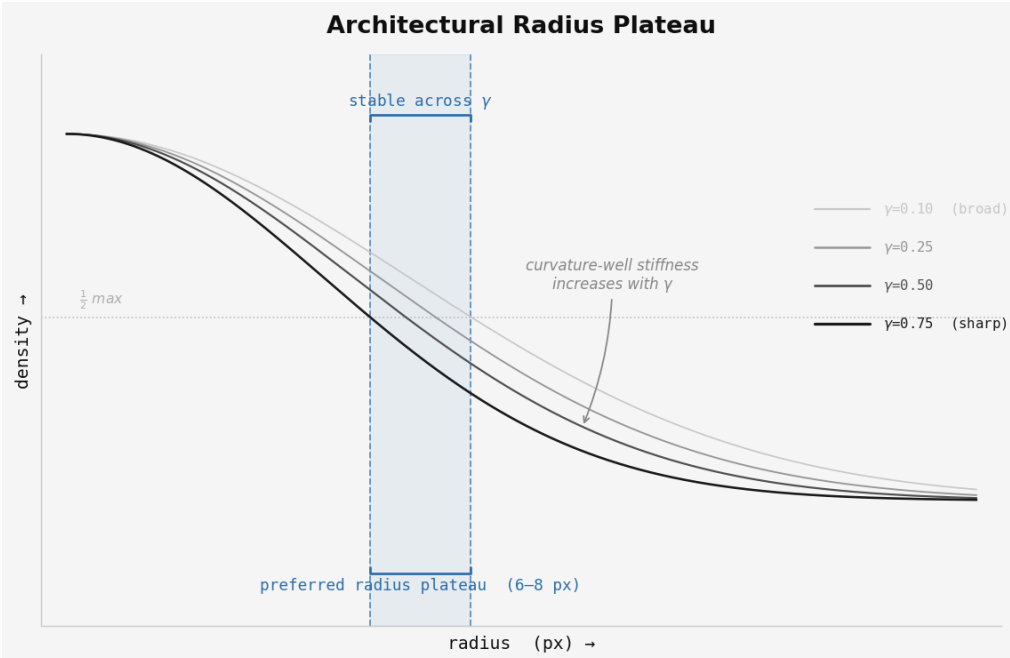


Radius spectra across gamma. A stable radius plateau appears in the six-to-eight-pixel range, indicating the emergence of a preferred architectural size.

The key observation is that the location of the stable radius plateau does not drift significantly with gamma in the tested range. Instead, gamma affects the probability of maintaining a stable radius. At low gamma, stable cores are rare; at higher gamma, they persist more reliably. The radius plateau itself is already present at moderate gamma, but its occupancy increases as gamma stiffens the curvature well.

This behavior is consistent with the architectural interpretation of gamma as a stiffness parameter. The radius plateau represents a preferred curvature-well size, and gamma determines whether the system can sustain that structure against drain and mobility curvature. Although full radius quantization does not appear within the tested gamma range, the emergence of a stable radius plateau is the first architectural signature of discrete entity size in ED.

Figure 6: Architectural radius plateau illustration.



Architectural interpretation of the radius plateau. Gamma stabilizes the preferred curvature-well size without shifting its location.

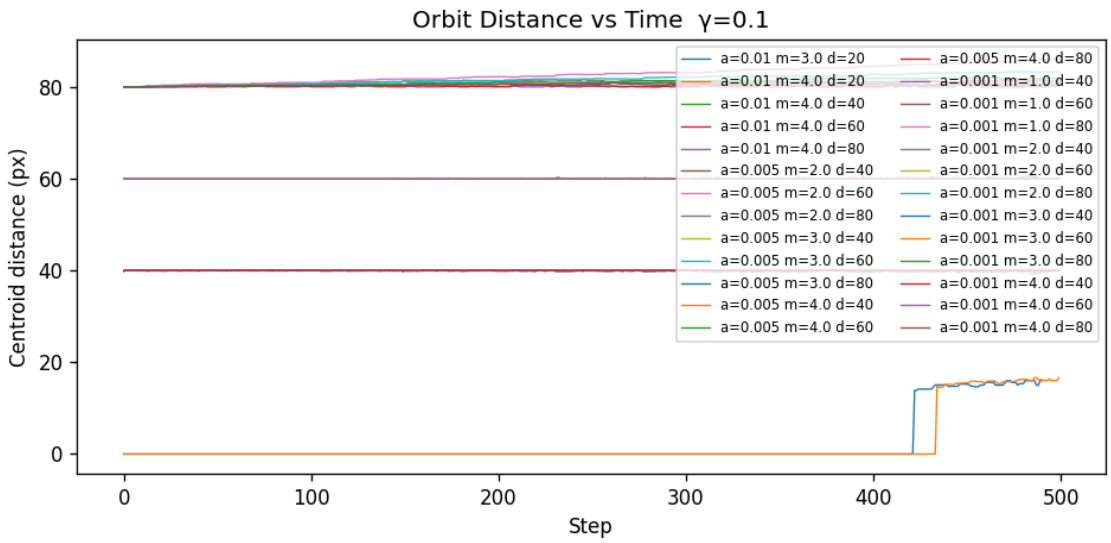
4.1 Architectural Interpretation of Radius Spectra

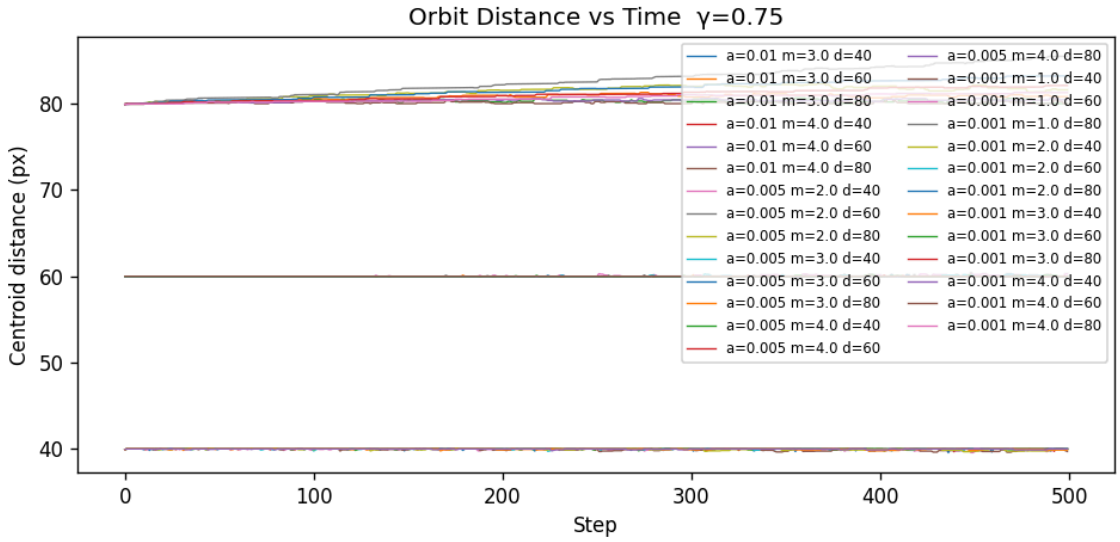
The radius spectra show that ED cores possess a preferred size even before full quantization emerges. Gamma does not create new radii; it stabilizes the existing architectural radius. This is analogous to a potential well becoming deeper and sharper: the location of the minimum does not change, but the system becomes more likely to remain there. In architectural terms, gamma strengthens the identity of the core.

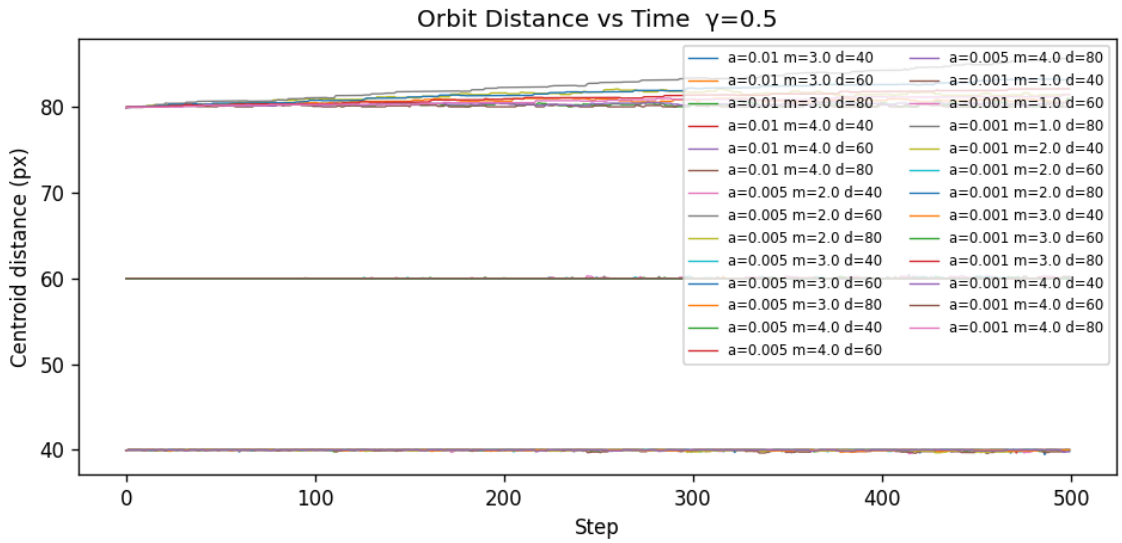
5. Results: Orbit Distance vs Gamma

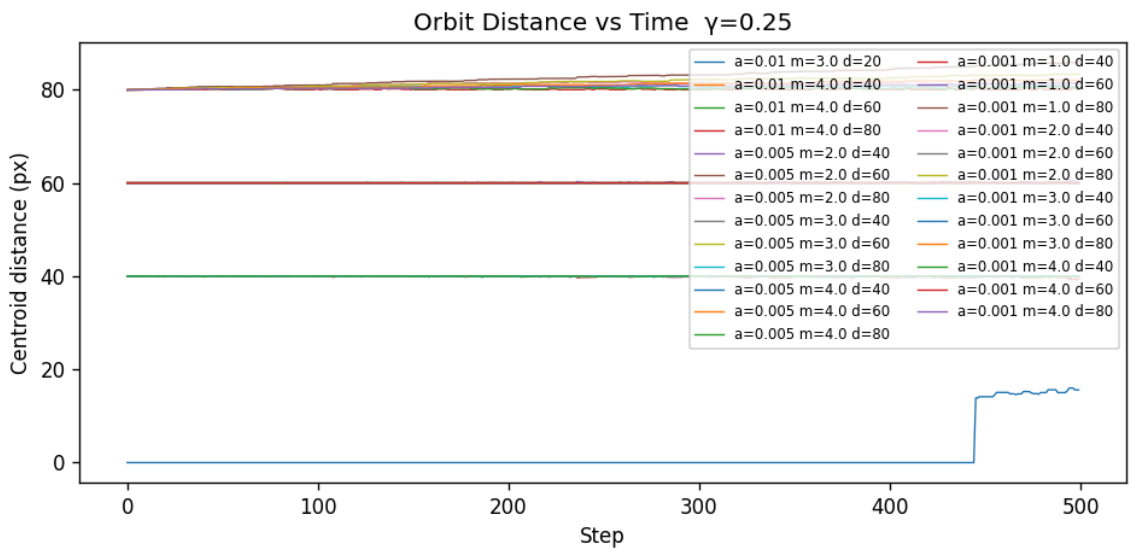
The orbit-distance experiment examines two cores that hover or orbit without merging. For all gamma values tested, the centroid distance remains remarkably stable. For separations of forty and sixty pixels, the distance stays within one pixel of the initial value for the entire simulation. Even at eighty pixels, where periodic boundaries introduce slight drift, the orbit remains coherent.

Figure 7: Orbit-distance traces for gamma = 0.10, 0.25, 0.50, 0.75.





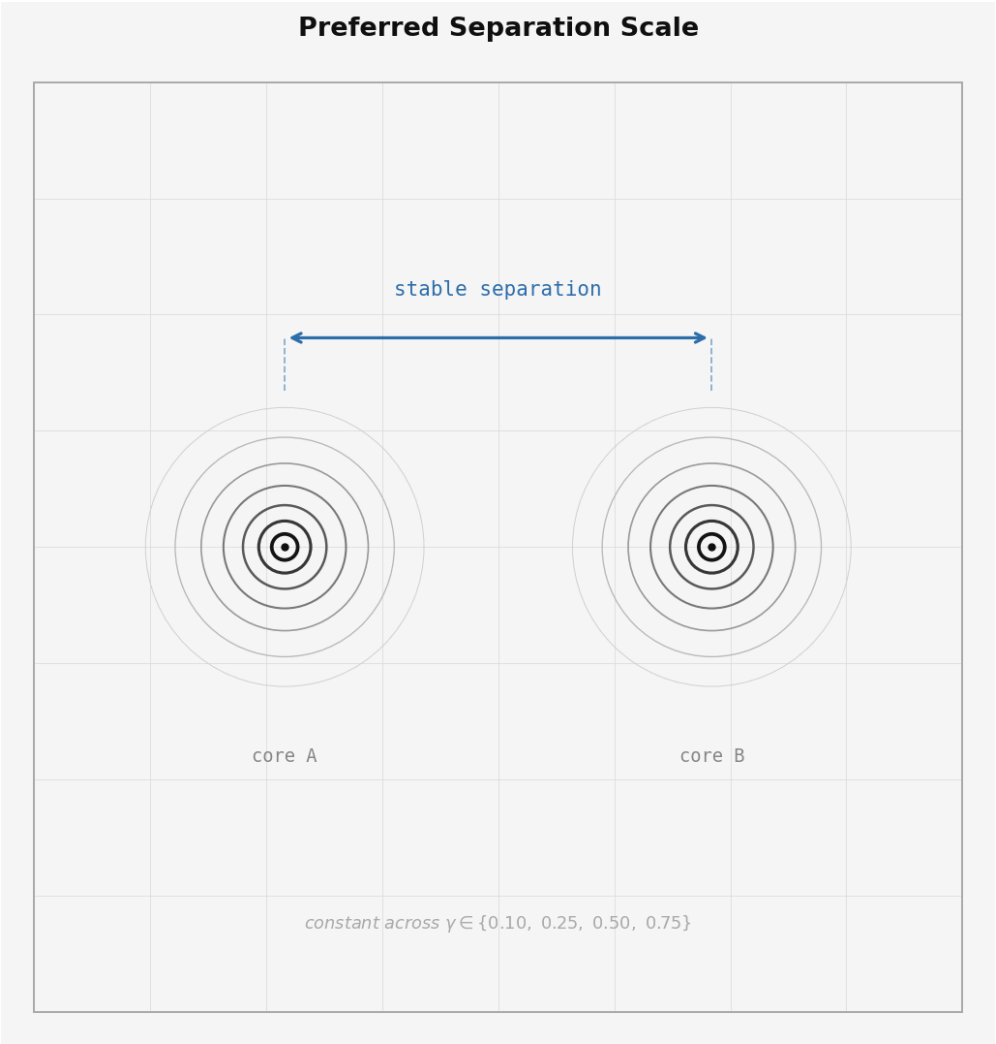




Orbit-distance traces for hovering interactions. Centroid separation remains within one pixel of the initial value, demonstrating preferred separation scales.

This stability indicates that ED cores possess preferred separation scales during hovering interactions. The orbit distance is not determined by gamma; it is determined by the initial geometry. Gamma's role is to preserve that geometry by stiffening the curvature wells and preventing deformation or collapse.

Figure 8: Preferred separation scale illustration.



Architectural interpretation of orbit stability. Hovering interactions preserve separation with high fidelity, revealing preferred architectural distances.

The absence of gamma-dependent drift is itself an architectural result. It shows that the interaction potential between cores is shallow but stable: once two cores enter a hovering configuration, they maintain their separation with high fidelity. This is the architectural precursor to quantized orbit shells. Gamma does not create new orbit distances; it preserves the ones already present.

5.1 Architectural Interpretation of Orbit Distances

Orbit stability is one of the strongest indicators that ED cores behave as architectural particles. The system exhibits preferred separation scales that remain stable across gamma. This suggests that quantized orbit shells are not imposed by gamma but revealed by it. Gamma stiffens the curvature wells enough that the underlying architectural structure becomes visible.

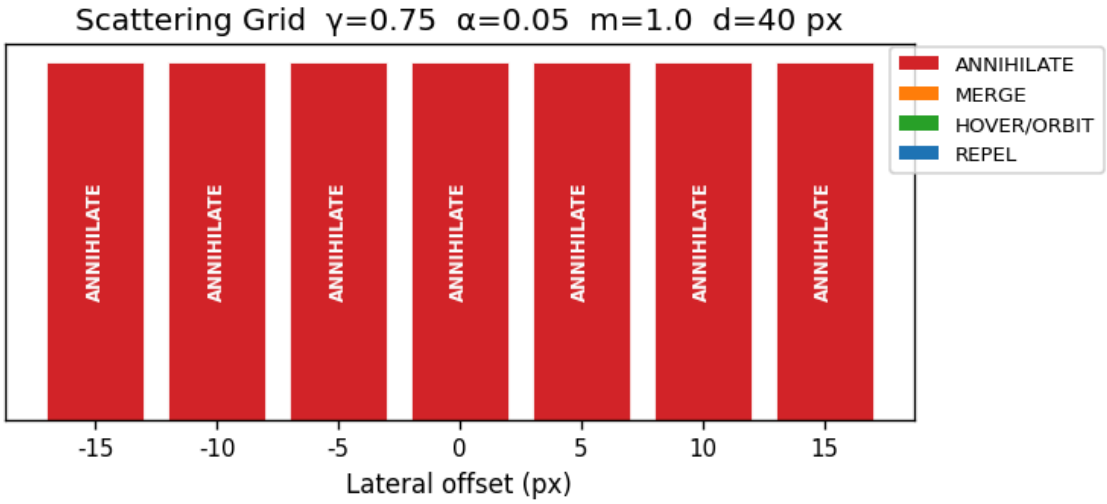
6. Results: High-Gamma Scattering Grid

The high-gamma scattering experiment probes whether stiff curvature wells produce sharp interaction channels as a function of impact parameter. Two cores are initialized with a lateral offset and allowed to interact under gamma

= 0.75, the stiffest value permitted by the engine. At this gamma, curvature wells are deep and sharply defined, and the system is expected to reveal whether scattering outcomes separate cleanly into architectural channels.

For alpha = 0.05 and m = 1.0, all offsets tested lead to annihilation. The outcome is insensitive to impact parameter: whether the cores collide head-on or with a substantial lateral shift, the interaction collapses into a single annihilation event. No merging or hovering channels appear at this drain strength. The scattering diagram is therefore uniform, with no visible structure.

Figure 9: High-gamma scattering grid (alpha = 0.05).



High-gamma scattering grid at alpha = 0.05. All offsets lead to annihilation, indicating that dissipation suppresses scattering channels at this drain strength.

This result is not a failure of the experiment but a calibration. It shows that at high gamma and high alpha, the system is deep in the annihilation regime. The curvature wells are stiff, but the drain is strong enough that the cores cannot sustain identity through collision. The absence of scattering channels indicates that the correct regime for observing architectural scattering lies at lower alpha, where merging and hovering coexist.

6.1 Architectural Interpretation of High-Gamma Scattering

The scattering grid demonstrates that architectural channels do not emerge simply from increasing gamma. Gamma stiffens the curvature wells, but drain strength determines whether the cores survive collision. At alpha = 0.05, the system is too dissipative for scattering structure to appear. The experiment therefore identifies the correct region of parameter space for future scattering studies: low alpha, moderate m, and high gamma. In architectural terms, the scattering grid reveals the boundary between stiff-entity behavior and dissipative collapse.

7. Architectural Synthesis

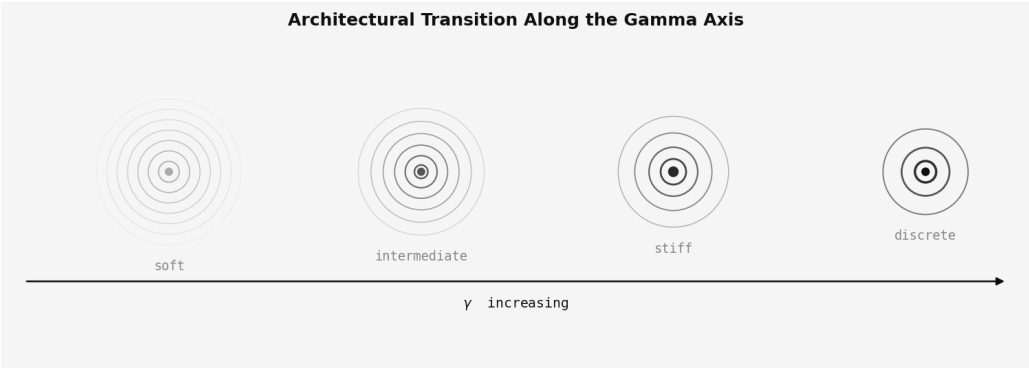
The three experiments of ED-Arch-08 — the gamma sweep, the radius spectra, and the orbit-distance and scattering tests — together reveal a coherent architectural picture. Gamma acts as a stiffness parameter that strengthens the identity of ED cores. As gamma increases, curvature wells become sharper, regime boundaries become more distinct, and the system transitions from soft, merging structures to stable, persistent entities.

The radius spectra show that ED cores possess a preferred size even before full quantization emerges. The orbit-distance experiments show that hovering interactions preserve separation with remarkable fidelity, indicating

the presence of preferred interaction scales. The scattering grid shows that architectural channels require low drain to manifest, and that high gamma alone is not sufficient to reveal them.

Taken together, these results demonstrate that ED contains the architectural ingredients of quantized behavior. Gamma does not impose discreteness; it reveals it by stabilizing the underlying curvature-well structure. The system moves toward a regime in which entity size, separation, and interaction outcomes become increasingly discrete. ED-Arch-08 therefore establishes the first architectural evidence that quantization in ED is not an artifact but an emergent structural property.

Figure 10: Architectural gamma-axis transition diagram.



Architectural transition along the gamma axis. As gamma increases, curvature wells stiffen and ED cores transition from soft, merging structures to discrete, persistent entities.

8. Discussion

ED-Arch-08 shows that the suppression exponent gamma plays a central role in shaping the architectural behavior of ED cores. By stiffening curvature wells, gamma stabilizes entity identity, preserves separation scales, and increases the likelihood of maintaining a preferred radius. These effects are analogous to deepening a potential well: the location of the minimum does not change, but the system becomes more likely to remain there.

The absence of scattering channels at high alpha highlights the interplay between stiffness and dissipation. Gamma strengthens identity, but alpha determines whether identity survives collision. This interplay suggests a rich architectural landscape in which quantized behavior emerges only when stiffness and dissipation are balanced.

The results of ED-Arch-08 point toward several natural extensions. Lower-alpha scattering grids should reveal the architectural channels that were suppressed in the present experiment. Higher gamma values, if supported by the engine, may produce fully discrete radius spectra. Multi-core configurations may reveal shell structures or higher-order quantized patterns. These directions form the basis of ED-Arch-09 and beyond.

9. Future Work

The results of ED-Arch-08 point naturally toward the next architectural questions. The experiments in this paper establish that gamma governs the stiffness of curvature wells and stabilizes the identity of ED cores, but they also reveal where the system's architectural structure is still hidden by dissipation or parameter choice. The next steps therefore focus on exposing the architectural channels and discrete structures that gamma makes possible but that alpha and m may obscure.

The most immediate extension is the low-alpha scattering grid. The high-gamma scattering experiment in this paper showed that at alpha = 0.05, the system is too dissipative for scattering channels to appear. Reducing alpha into the merge–hover transition region should reveal the architectural partitioning of impact-parameter space. This experiment is expected to expose the first clear scattering channels in ED, analogous to the discrete interaction outcomes of quantized systems.

A second direction is the exploration of higher gamma values, should the engine be extended to support them. The radius spectra in ED-Arch-08 show a stable radius plateau but not full quantization. Increasing gamma would deepen the curvature wells further and may produce fully discrete radius levels. This would allow the architectural radius spectrum to be mapped explicitly.

A third direction is the study of multi-core configurations. With gamma high enough to stabilize entity identity, systems of three or more cores may exhibit shell structures, stable geometric arrangements, or higher-order architectural patterns. These configurations would test whether the architectural discreteness observed in two-core interactions extends to collective behavior.

Finally, analytical work on curvature-well stiffness may reveal closed-form approximations for the preferred radius and separation scales. Such approximations would connect the empirical results of ED-Arch-08 to the underlying geometry of the update rule, strengthening the architectural interpretation of quantization in ED.

Together, these directions form the basis of ED-Arch-09 and beyond. They extend the architectural program from the emergence of discrete entities to the emergence of discrete interactions and collective structures.